

Program 5: INTEGRATION BY RUNNING JENKINS IN A DOCKER CONTAINER

Step1: ensure Docker service is running

>docker- version

>docker compose version

Step2: mkdir program5

Cd prgram5

Step3:

Create a dockerfile with the code

```
FROM jenkins/jenkins:lts
#we have to set root as the user
USER root
#we have to update and install the docker input and output
RUN apt-get update && apt-get install -y docker.io
#we have to change the user to jenkins
USER jenkins
```

Step4:

Create a file called docker-compose.yml

```
version: '3.8'
services:
  jenkins:
    image: jenkins/jenkins:lts
    container_name: jenkins
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
volumes:
  jenkins_home:
```

Step5: build an image

>docker build-t program5 .

Docker run the container

>docker run -p 8080:8080 program5

```
RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/program5$ docker build -t program5 .
[+] Building 244.6s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 272B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407dfa9bcb122cc4d71d4ce44d2e3b2be31af004c157453a935
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407dfa9bcb122cc4d71d4ce44d2e3b2be31af004c157453a935
=> => sha256:1e407dfa9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 3.12kB / 3.12kB
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa812cc605644 12.71kB / 12.71kB
=> => sha256:b768bb669b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e893b800 2.58kB / 2.58kB
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB
=> => sha256:5687d4df0728e358596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 65.08MB / 65.08MB
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627 95.32MB / 95.32MB
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 189B / 189B
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acb06245 6.29MB / 6.29MB
=> => sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bfd6899 72.05MB / 72.05MB
=> => extracting sha256:5687d4df0728e358596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc
=> => sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de53927 2.23kB / 2.23kB
=> => extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471
=> => extracting sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5
=> => extracting sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168
=> => sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbeea 1.29kB / 1.29kB
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 392B / 392B
=> => extracting sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627
=> => extracting sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98
=> => extracting sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acb06245
=> => extracting sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bfd6899
=> => extracting sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de53927
=> => extracting sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbeea
=> => extracting sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313
=> [2/2] RUN apt-get update && apt-get install -y docker.io
=> exporting to image
=> => exporting layers
=> => writing image sha256:5cb86b3c9bfe5be89122bc9ba9d52ad97fee44298766385b444a7420f89c8c28
=> => naming to docker.io/library/program5
```

```

jvm 21.0.9+10-LTS
!025-12-29 06:50:23.914+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty
!025-12-29 06:50:23.949+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
!025-12-29 06:50:24.137+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home drectory: /var/jenkins_home found at: EnvVars.master
!025-12-29 06:50:24.197+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3c782d8e{Jenki
is_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9{STARTED}}
!025-12-29 06:50:24.203+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac{HTTP/1.1, (http/1.1)}{0.0.0.0:8
!025-12-29 06:50:24.211+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4{STARTING}[12.0.25,sto=0] @1399ms
!025-12-29 06:50:24.211+0000 [id=43] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
!025-12-29 06:50:24.288+0000 [id=42] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.3
!025-12-29 06:50:24.344+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
!025-12-29 06:50:24.357+0000 [id=72] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
!025-12-29 06:50:24.758+0000 [id=72] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
!025-12-29 06:50:24.762+0000 [id=56] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
!025-12-29 06:50:24.763+0000 [id=71] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
!025-12-29 06:50:24.855+0000 [id=52] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
!025-12-29 06:50:24.855+0000 [id=62] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
!025-12-29 06:50:24.855+0000 [id=72] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
!025-12-29 06:50:24.857+0000 [id=70] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
!025-12-29 06:50:24.867+0000 [id=86] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
!025-12-29 06:50:25.101+0000 [id=63] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> c1a7f769dad74846b06f4d24884607a4
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
!025-12-29 06:50:45.396+0000 [id=63] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
!025-12-29 06:50:45.417+0000 [id=42] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
!025-12-29 06:50:47.497+0000 [id=86] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstal
!025-12-29 06:50:47.498+0000 [id=86] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
]

```

Check it in
><http://localhost:8080>

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

`/var/jenkins_home/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

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